Compositional modeling

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Introduction

The central idea of algebraicodin is **compositional modeling**: build complex models by assembling reusable components. Instead of writing a monolithic SIR model, we define *infection* and *recovery* as separate **open Petri nets**, then compose them by declaring how they share species.

This approach mirrors the mathematical framework of AlgebraicPetri.jl and uses **undirected wiring diagrams** (UWDs) from applied category theory to specify the composition pattern.

```
library(algebraicodin)
```

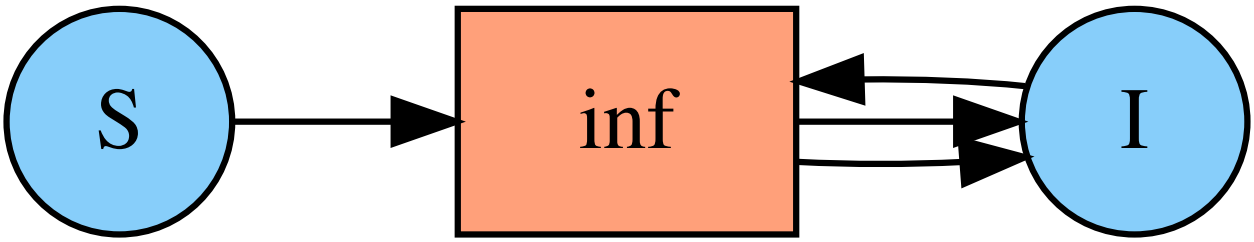
Building blocks

An **open Petri net** is a Petri net with designated *legs*—species that are exposed for composition. The package provides helpers for common epidemiological transitions:

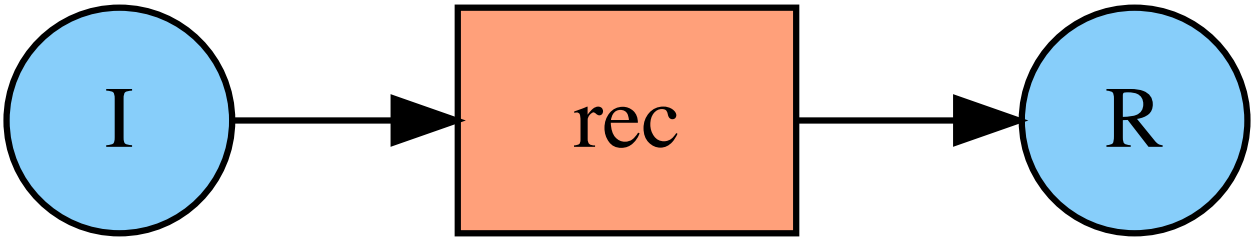
```
# Infection: S + I → I + I (frequency-dependent transmission)
infection <- exposure_petri("S", "I", "I", "inf")

# Recovery: I → R (spontaneous transition)
recovery <- spontaneous_petri("I", "R", "rec")
```

Each component is self-contained with its own species and transitions. The legs expose species that will be identified during composition.

```
plot_petri(infection)
```




```
plot_petri(recovery)
```



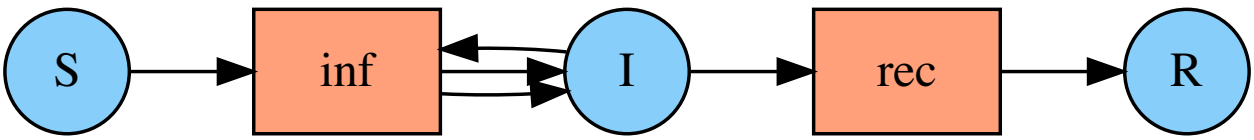
Composing with the %o% operator

The simplest way to compose two open Petri nets is the `%o%` operator, which automatically identifies species with matching names:

```
sir_composed <- infection %o% recovery
species_names(apex(sir_composed))
#> [1] "S" "I" "R"
transition_names(apex(sir_composed))
#> [1] "inf" "rec"
```

Species "I" appears in both components, so it gets identified (shared). The result is the standard SIR model.

```
plot_petri(sir_composed)
```




```
sir_code <- pn_to_odin(apex(sir_composed), "ode")
cat(sir_code)
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf <- parameter()
#> rec <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Transition rates
#> rate_inf <- inf * S * I
#> rate_rec <- rec * I
#>
#> ## Derivatives
#> deriv(S) <- -rate_inf
#> deriv(I) <- rate_inf - rate_rec
#> deriv(R) <- rate_rec
```

Composing with wiring diagrams

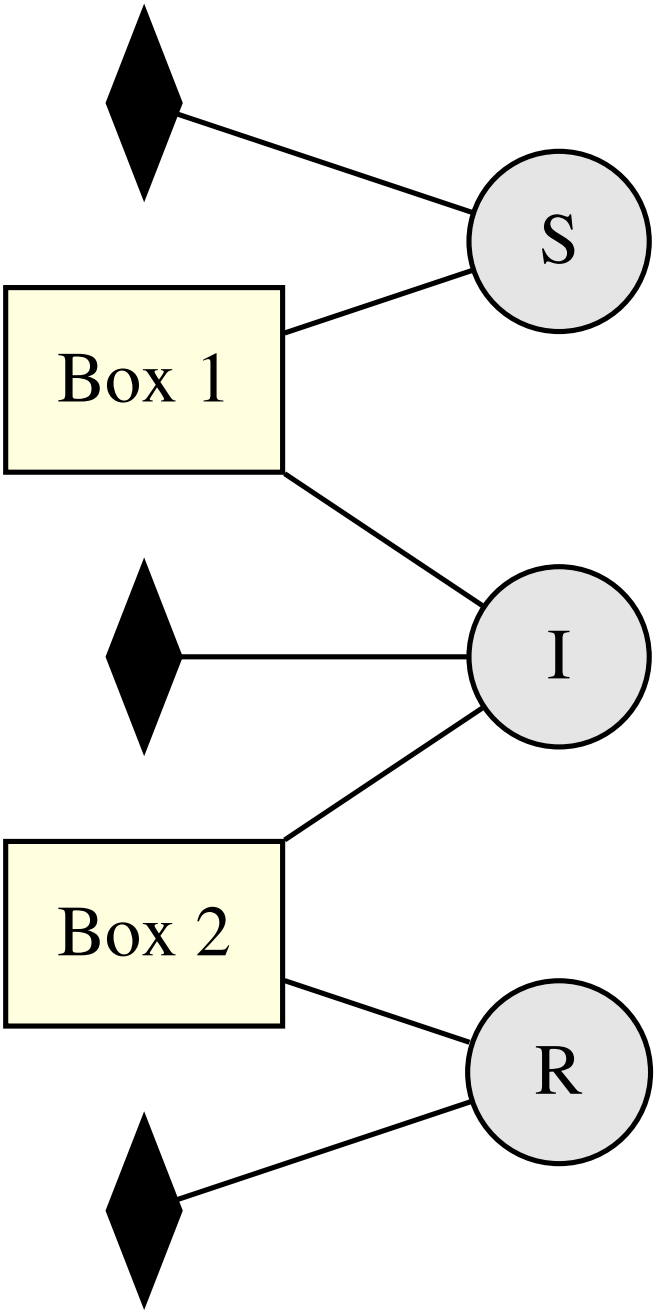
For more control, define an explicit **undirected wiring diagram** (UWD) and use `oapply()`. The UWD specifies *junctions* (shared species) and *boxes* (components):

```
w_sir <- catlab::uwd(
  outer = c("S", "I", "R"),
  infection = c("S", "I"),
  recovery  = c("I", "R")
)
sir_via_uwd <- oapply(w_sir, list(infection, recovery))
```

```r
species_names(apex(sir_via_uwd))
#> [1] "S" "I" "R"
```

The UWD approach is equivalent to `%o%` for two components but generalises naturally to more complex diagrams.

```r
plot_uwd(w_sir)
```



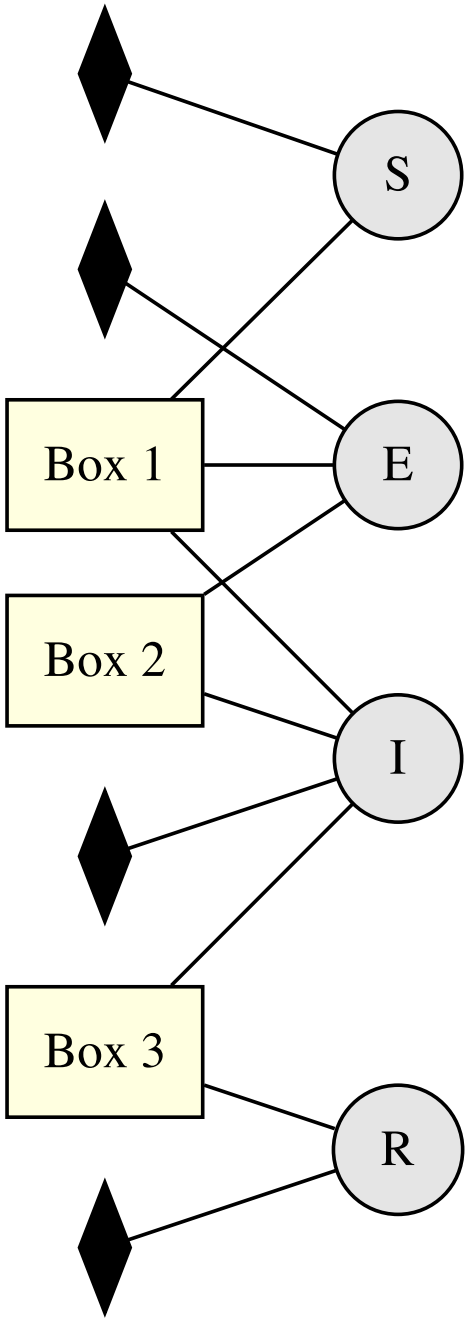
Extending to SEIR

Adding an exposed compartment requires a third component—*progression* from E to I:

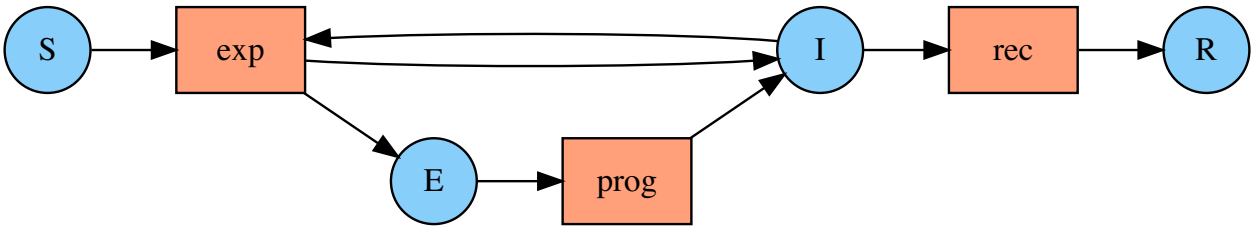
```r
# Exposure: S + I → E + I (infectious contact produces exposed, not infectious)
exposure <- exposure_petri("S", "I", "E", "exp")

# Progression: E → I
progression <- spontaneous_petri("E", "I", "prog")

# Compose with explicit UWD
w_seir <- catlab::uwd(
  outer    = c("S", "E", "I", "R"),
  exposure = c("S", "I", "E"),
  progression  = c("E", "I"),
  recovery = c("I", "R")
)
seir <- oapply(w_seir, list(exposure, progression, recovery))
species_names(apex(seir))
#> [1] "S" "E" "I" "R"
transition_names(apex(seir))
#> [1] "exp"  "prog" "rec"
```

The wiring diagram shows how the three components share species:

```r
plot_uwd(w_seir)
```




```r
plot_petri(seir)
```




```r
cat(pn_to_odin(apex(seir), "ode"))
#> ## Auto-generated by algebraicodin
#>
```

```
#> ## Parameters
#> exp <- parameter()
#> prog <- parameter()
#> rec <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> E0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(E) <- E0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Transition rates
#> rate_exp <- exp * S * I
#> rate_prog <- prog * E
#> rate_rec <- rec * I
#>
#> ## Derivatives
#> deriv(S) <- -rate_exp
#> deriv(E) <- rate_exp - rate_prog
#> deriv(I) <- rate_prog - rate_rec
#> deriv(R) <- rate_rec
```

Comparison: composed vs hand-written SEIR

The hand-written odin2 SEIR model:

```
seir_handwritten <- "
exp <- parameter()
prog <- parameter()
rec <- parameter()
S0 <- parameter()
E0 <- parameter()
I0 <- parameter()
R0 <- parameter()
initial(S) <- S0
initial(E) <- E0
initial(I) <- I0
initial(R) <- R0
rate_exp <- exp * S * I
rate_prog <- prog * E
rate_rec <- rec * I
deriv(S) <- -rate_exp
deriv(E) <- rate_exp - rate_prog
deriv(I) <- rate_prog - rate_rec
deriv(R) <- rate_rec
"
```

Let us run both and verify they produce identical trajectories:

```r
# Algebraic version
gen_alg <- odin2::odin(pn_to_odin(apex(seir), "ode"))
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 13 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.system523a0040
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.system523a0040’ ...
#>    ** this is package ‘odin.system523a0040’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:92:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                           ^
#>    dust.cpp:96:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>     96 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_n_
#>          |                     ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpMjuTY0/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system523a0040) (376ms)
#>
#> i Loading odin.system523a0040
pars <- list(exp = 0.001, prog = 0.2, rec = 0.1,
             S0 = 990, E0 = 0, I0 = 10, R0 = 0)
sys_alg <- dust2::dust_system_create(gen_alg, pars, n_particles = 1)
dust2::dust_system_set_state_initial(sys_alg)
t <- seq(0, 100, by = 0.5)
y_alg <- dust2::dust_system_simulate(sys_alg, t)

# Hand-written version
gen_hw <- odin2::odin(seir_handwritten)
#> i Using cached generator
sys_hw <- dust2::dust_system_create(gen_hw, pars, n_particles = 1)
dust2::dust_system_set_state_initial(sys_hw)
```

```r
y_hw <- dust2::dust_system_simulate(sys_hw, t)

cat("Max absolute difference:", max(abs(y_alg - y_hw)), "\n")
#> Max absolute difference: 0
```

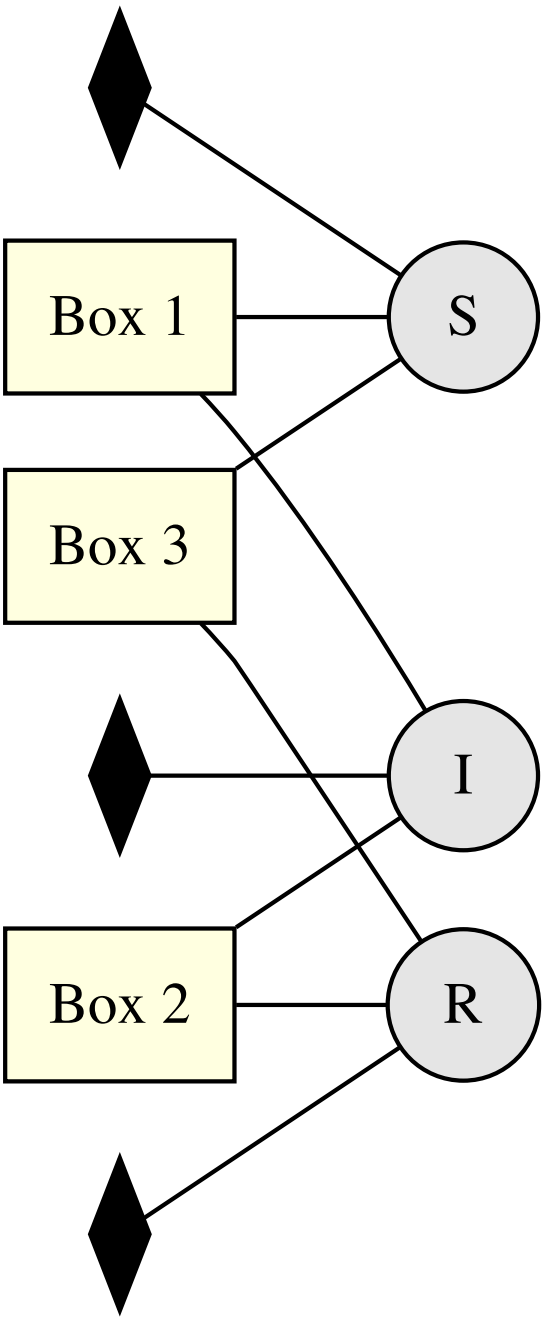
The outputs match exactly—both compile to the same C++ system.

Adding vaccination: SIRV

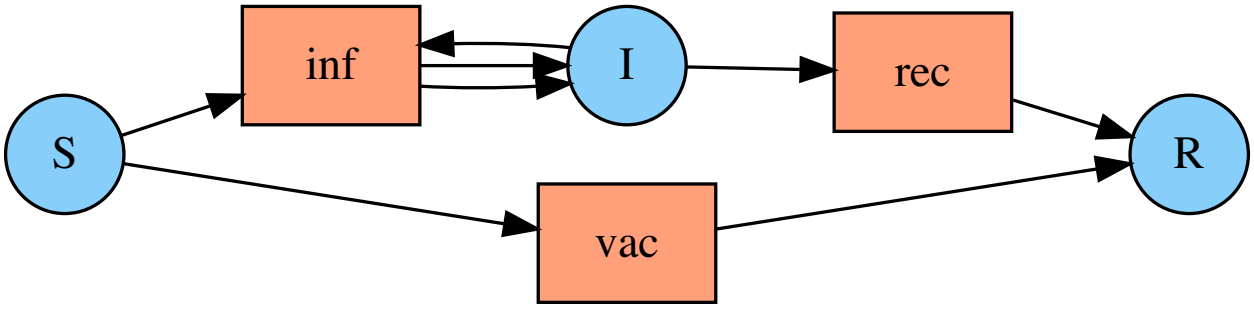
We can extend SIR with a vaccination transition (S → R at rate `vac`):

```r
vaccination <- spontaneous_petri("S", "R", "vac")

w_sirv <- catlab::uwd(
  outer      = c("S", "I", "R"),
  infection  = c("S", "I"),
  recovery   = c("I", "R"),
  vaccination = c("S", "R")
)
sirv <- oapply(w_sirv, list(infection, recovery, vaccination))
cat(pn_to_odin(apex(sirv), "ode"))
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf <- parameter()
#> rec <- parameter()
#> vac <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Transition rates
#> rate_inf <- inf * S * I
#> rate_rec <- rec * I
#> rate_vac <- vac * S
#>
#> ## Derivatives
#> deriv(S) <- -rate_inf - rate_vac
#> deriv(I) <- rate_inf - rate_rec
#> deriv(R) <- rate_rec + rate_vac
```

```r
plot_uwd(w_sirv)
```

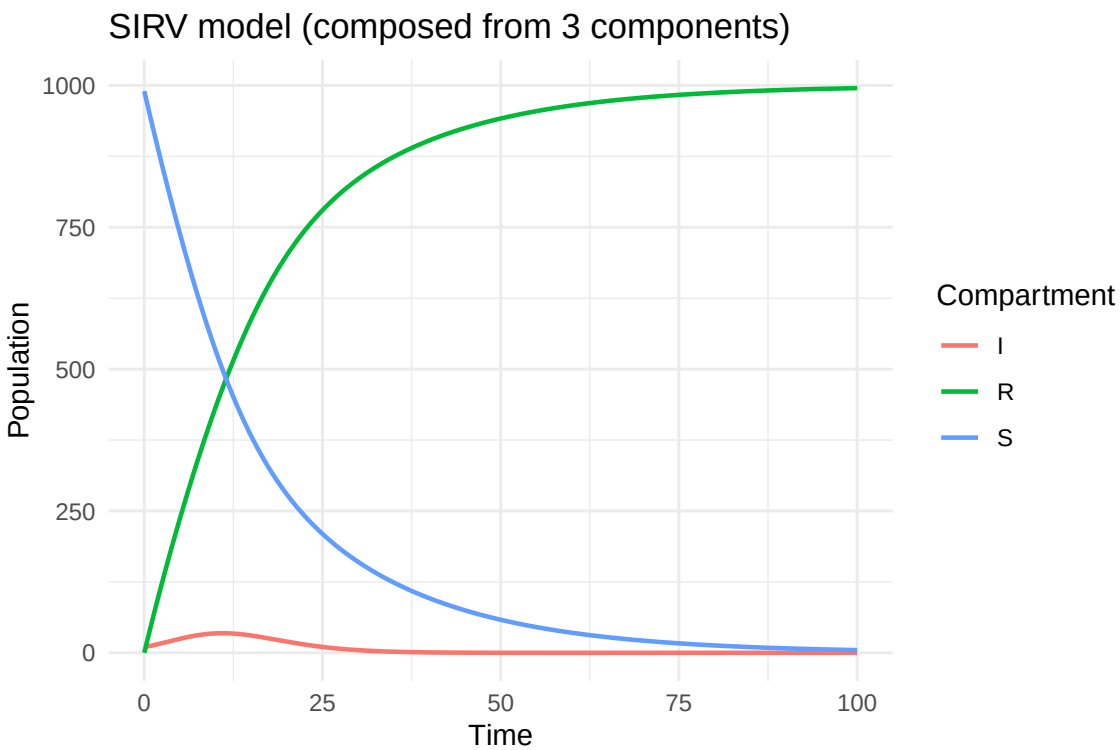



```r
plot_petri(sirv)
```



```r
gen <- odin2::odin(pn_to_odin(apex(sirv), "ode"))
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 13 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.systemd0eef7b9
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.systemd0eef7b9’ ...
#>    ** this is package ‘odin.systemd0eef7b9’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:84:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                           ^
#>    dust.cpp:88:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>       88 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_
#>          |                     ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpMjuTY0/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.systemd0eef7b9) (450ms)
#>
#> i Loading odin.systemd0eef7b9
pars <- list(inf = 0.0005, rec = 0.25, vac = 0.05,
             S0 = 990, I0 = 10, R0 = 0)
sys <- dust2::dust_system_create(gen, pars, n_particles = 1)
dust2::dust_system_set_state_initial(sys)
y <- dust2::dust_system_simulate(sys, t)

df <- data.frame(
  time = rep(t, 3),
  value = c(y[1, ], y[2, ], y[3, ]),
  compartment = rep(c("S", "I", "R"), each = length(t))
)
library(ggplot2)
ggplot(df, aes(time, value, colour = compartment)) +
```

```
  geom_line(linewidth = 0.8) +
  labs(title = "SIRV model (composed from 3 components)",
       x = "Time", y = "Population", colour = "Compartment") +
  theme_minimal()
```



Using the epidemiology dictionary

For convenience, `epi_dict()` provides pre-built components for common transitions:

```
dict <- epi_dict()
names(dict)
#> [1] "infection"       "recovery"        "vaccination"     "waning"
#> [5] "death"           "exposure"        "progression"     "hospitalization"
```

You can compose a model directly from the dictionary using `compose_epi()`:

```
w <- catlab::uwd(
  outer     = c("S", "I", "R"),
  infection = c("S", "I"),
  recovery  = c("I", "R")
)
sir <- compose_epi(w, dict, c("infection", "recovery"))
species_names(apex(sir))
#> [1] "S" "I" "R"
```

Composing more than two components

The `compose()` function chains multiple `%o%` operations:

```
sirh <- compose(
  exposure_petri("S", "I", "I", "inf"),
```

```r
  spontaneous_petri("I", "R", "rec"),
  spontaneous_petri("I", "H", "hosp"),
  spontaneous_petri("H", "R", "hosp_rec")
)
species_names(apex(sirh))
#> [1] "S" "I" "R" "H"
transition_names(apex(sirh))
#> [1] "inf"      "rec"      "hosp"     "hosp_rec"
```

Summary

Method	Syntax	Best for
`%o%`	`a %o% b`	Quick binary composition
`oapply()`	`oapply(uwd, components)`	Full control via wiring diagrams
`compose()`	`compose(a, b, c)`	Chaining multiple components
`compose_epi()`	`compose_epi(uwd, dict, names)`	Dictionary-based workflows

The compositional approach ensures that every shared species is correctly identified and that stoichiometry is preserved across components. For stratification by age, risk, or vaccination status, see `vignette("stratification")`.